Algebra & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Name, identify, order, and compare fractions

Rationalise the denominator and simplify: $\frac{10}{3-\sqrt{5}}$.

2 [Foundational] • Target: Equivalent and simplified fractions

A bag contains 6 green counters and 4 yellow counters. A counter is chosen at random, its colour recorded, and it is REPLACED. A second counter is then chosen. What is the probability that both counters are the same colour?

3 [Foundational] • Target: Adding and subtracting fractions

A car travels a distance of 150 km, correct to the nearest 10 km, in a time of 2 hours, correct to the nearest 0.1 hour. What is the upper bound of the average speed in km/h?

4 [Foundational] • Target: Fractions of a quantity

In a survey of 50 tourists, 30 visited museum M, 25 visited palace P, and 20 visited art gallery G. 12 visited both M and P, 10 visited both P and G, 8 visited both M and G, and 5 visited all three. How many tourists visited NONE of the three attractions?

5 [Foundational] • Target: Multiplying fractions

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

6 [Foundational] • Target: Fractions and division

Port B is 40 km from Port A on a bearing of 120°. Port C is 50 km from Port A on a bearing of 030°. Calculate the distance between Port B and Port C.

7 [Foundational] • Target: Name, identify, order, and compare fractions

A drawer contains 5 blue socks, 4 grey socks, and 3 black socks. Two socks are pulled at random without replacement. What is the probability of pulling two socks of different colours?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Equivalent and simplified fractions

Convert $0.1\dot{4}\dot{8}$ to a fraction in its simplest form.

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9 [Intermediate] • Target: Adding and subtracting fractions

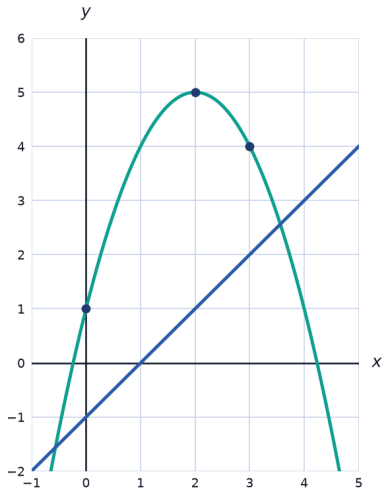
In a circle of radius 10 cm, an arc subtends a central angle of 72°. Calculate the perimeter of this sector to 1 decimal place.

10 [Intermediate] • Target: Fractions of a quantity

A vertical pole of height 12 m casts a shadow of length 9 m on horizontal ground. Calculate the angle of elevation of the sun to the nearest degree.

11 [Intermediate] • Target: Adding and subtracting fractions

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(5)$.

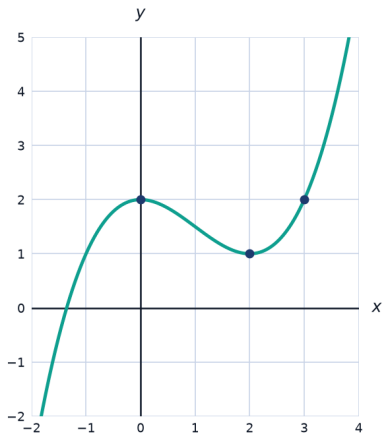


12 [Intermediate] • Target: Adding and subtracting fractions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

13 [Intermediate] • Target: Fractions of a quantity

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(2)$.



14 [Intermediate] • Target: Fractions of a quantity

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Fractions of a quantity** • **Cross-Lesson: 8.01 Probability & Statistics**

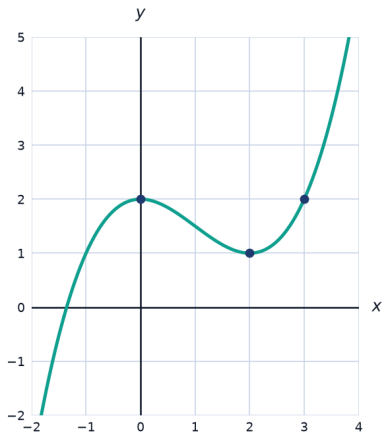
A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

16 **[Challenging]** • **Target: Fractions of a quantity** • **Cross-Lesson: 9.01 Functions**

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(3)$.

17 **[Challenging]** • **Target: Fractions of a quantity** • **Cross-Lesson: 1.01 Number**

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



18 **[Challenging]** • **Target: Fractions of a quantity** • **Cross-Lesson: 10.01 Sets & Venn Diagrams**

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(4)$.

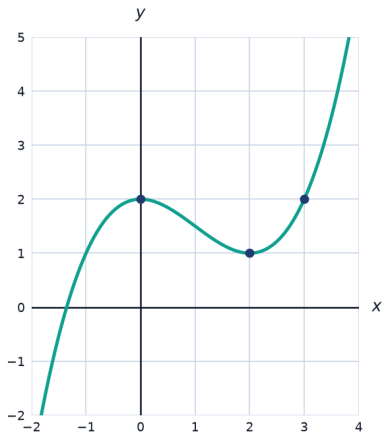
19 **[Challenging]** • **Target: Fractions of a quantity** • **Cross-Lesson: 3.01 Coordinate Geometry**

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

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20 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 4.01 Mensuration

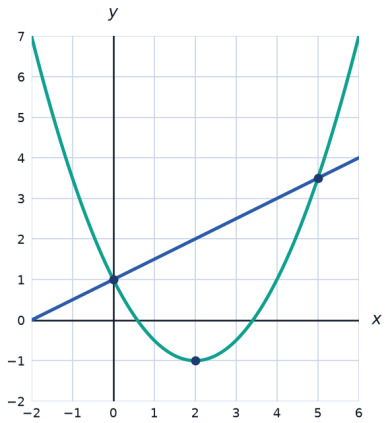
For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(5)$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(2)$.



a. 16

b. 20

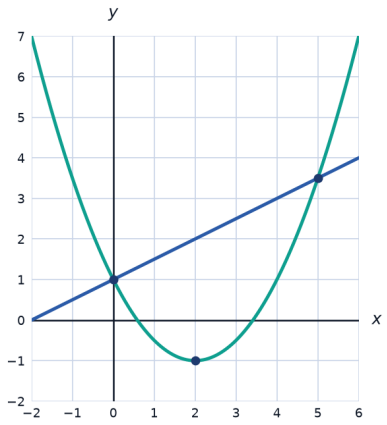
c. 8

d. 20

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22 [Foundational] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

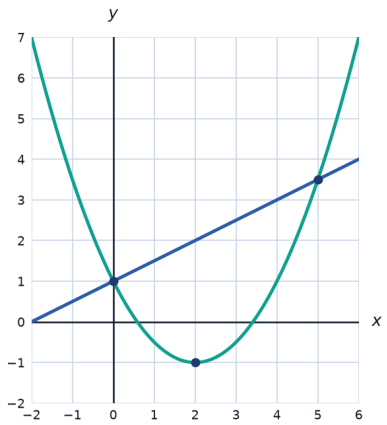
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

23 [Foundational] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(5)$.



a. 28

b. 32

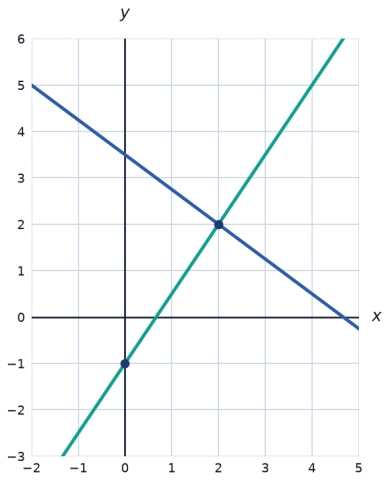
c. 20

d. 32

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24 [Foundational] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(2)$.



a. 19

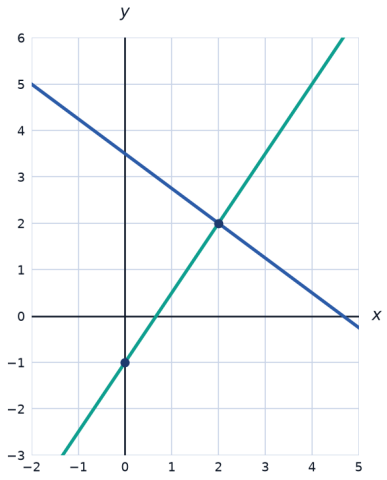
b. 24

c. 10

d. 23

25 [Foundational] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

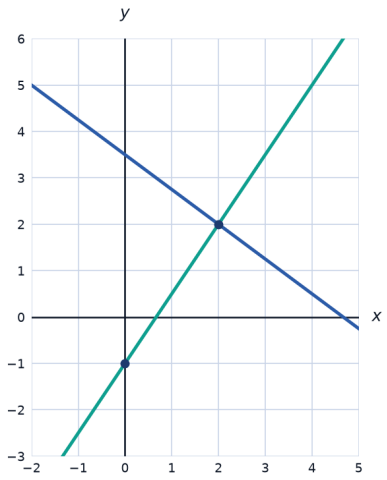
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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26 [Foundational] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(5)$.



a. 34

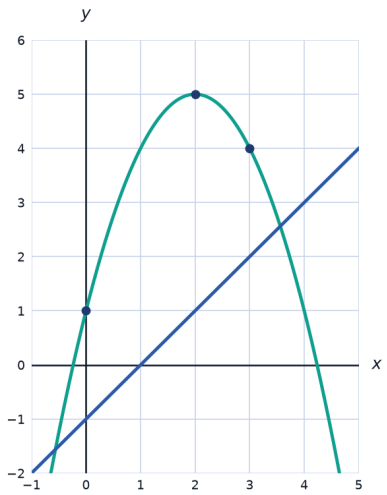
b. 39

c. 25

d. 38

27 [Foundational] • Target: Adding and subtracting fractions

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(2)$.



a. 15

b. 21

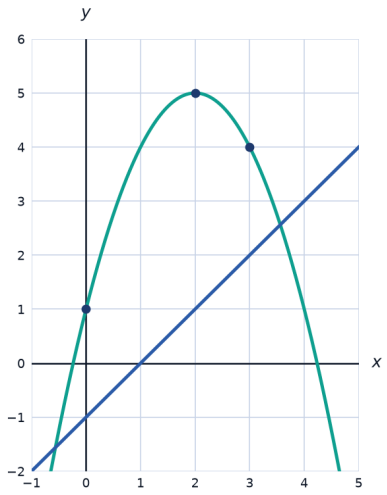
c. 12

d. 19

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28 [Foundational] • Target: Adding and subtracting fractions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Name, identify, order, and compare fractions

A data set recorded for name, identify, order, and compare fractions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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31 [Intermediate] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(3)$.

a. 20

b. 24

c. 12

d. 24

32 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

33 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

34 [Intermediate] • Target: Equivalent and simplified fractions

A data set recorded for equivalent and simplified fractions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

35 [Intermediate] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(3)$.

a. 24

b. 29

c. 15

d. 28

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36 [Intermediate] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

37 [Intermediate] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 10.01 Sets & Venn Diagrams

A data set recorded for adding and subtracting fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

39 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(3)$.

a. 21

b. 27

c. 18

d. 25

40 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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41 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 5.01 Geometry

A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

42 [Challenging] • Target: Multiplying fractions • Cross-Lesson: 6.01 Trigonometry

A data set recorded for multiplying fractions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

43 [Challenging] • Target: Multiplying fractions • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 3x + 5$ in multiplying fractions, evaluate $f(3)$.

a. 14

b. 17

c. 9

d. 18

44 [Challenging] • Target: Fractions and division • Cross-Lesson: 8.01 Probability & Statistics

A data set recorded for fractions and division consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

45 [Challenging] • Target: Fractions and division • Cross-Lesson: 9.01 Functions

For the model $f(x) = 4x + 6$ in fractions and division, evaluate $f(3)$.

a. 18

b. 22

c. 12

d. 22

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